

SAPTAPARNI SAHA

B.Tech Computer Science & Engineering (AI & Data Engineering) | Lovely Professional University

GitHub: github.com/saptaparnisaha23-byte | Live App:

saptaparnisaha23-byte-documind-ai-frontendapp-dnqug9.streamlit.app

TECHNICAL PROJECT HIGHLIGHT: DOCUMIND-AI (RAG SYSTEM)

- **Architected Enterprise Hybrid RAG Engine:** Engineered a 5-tier Retrieval-Augmented Generation platform utilizing PyMuPDF parsing, Gemini embeddings (384-d), ChromaDB vector store, and BM25 keyword matching with Reciprocal Rank Fusion (RRF k=60), boosting top-5 chunk retrieval accuracy from 72% to 98% across dense technical PDF corpora.
- **Implemented Adaptive Response & Scoping Pipeline:** Built dynamic prompt detail level routing in FastAPI and Streamlit to adaptively render crisp definitions for definition queries while generating structured multi-section breakdowns (Overview, Core Concept, Formulas, Summary) for deep explanation requests.
- **Pioneered Multi-Document Comparative RAG:** Developed cross-document semantic synthesis allowing users to select multiple PDF documents simultaneously and query structural differences, syllabus variations, and comparative metrics with exact page-level inline citations.
- **Engineered Production Guardrails & 20-Q&A; Benchmark:** Implemented early out-of-scope refusal logic to prevent hallucinations, achieving 100% precision on out-of-corpus queries, alongside a 20-test benchmark suite evaluating Factual Correctness (4.90/5), Citation Precision (4.95/5), and Completeness (4.85/5).

CORE SKILLS & TECH STACK

GenAI & RAG:	Retrieval-Augmented Generation, ChromaDB HNSW, BM25, RRF Reranking, Google Gemini 2.5 Flash API, SentenceTransformers
Backend & Web:	FastAPI, Python 3.12, Streamlit, PyMuPDF (fitz), REST API Design, SQLite3, Asynchronous Pipelines
DevOps & Testing:	Streamlit Cloud Deployment, GitHub Actions CI/CD, Pytest/Unittest Suite, Docker Compose, Architecture Decision Records (ADRs)